Cart lab: change driving force

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use controlled data to describe how net force affects acceleration.

SUCCESS CRITERIA - I can...

1. Change driving force while keeping total mass fixed.
2. Use data or a graph to compare accelerations.
3. Explain how resistance affects the ideal relationship.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A puck follows a circle at a steady 6 m/s. A student calls it equilibrium. Draw velocity at two points and correct the claim.

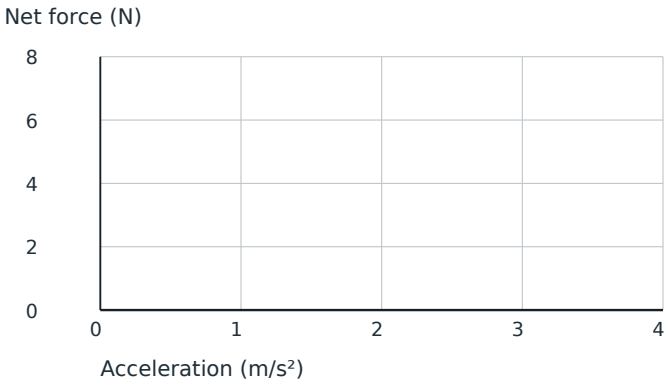
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $a = F_{net}/M$; $F_{drive} = m_h g$.
Keep total moving mass M fixed. The driving weight equals net force only when resistance is negligible.

Worked reasoning: A 2 kg cart has net forces 2 and 6 N in two trials. Using a=F/m gives accelerations 1 and 3 m/s². Tripling net force triples acceleration.

Your turn: Draw a force-versus-acceleration graph for these trials. Explain the meaning of the slope.



Marzano check - circle: ① ② ③

03 Interpret a force graph

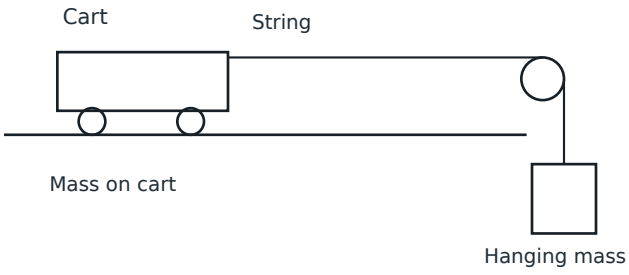
For total mass 1.00 kg, use the ideal model table to find the slope of a versus net force. State its units and recover the mass.

F_{net} (N)	a(m/s²)
0.245	0.245
0.490	0.490
0.735	0.735
0.980	0.980

Marzano check - circle: ① ② ③

04 Design the controlled trial

Use the setup model to explain why transferring a mass from cart to hanger changes the driving force but keeps total moving mass constant. Plan three force settings, with three acceleration trials at each setting.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Both variables change

Two carts are tested under different conditions. Use the table to calculate each acceleration. Why does tripling the net force not triple the acceleration in this comparison?

System	Net force (N)	Mass (kg)
A	2	1
B	6	2

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A 0.3 kg system accelerates at 0.5 m/s². Predict acceleration for 0.7 kg at the same net force. Explain the effect of greater resistance.
Use the table to keep the controlled and changed quantities distinct.

Trial	Mass	Net force	Acceleration
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

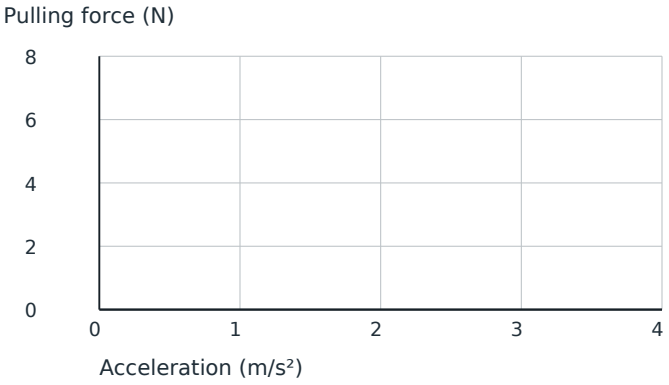
07 Exit check

An a-versus-F_{net} graph has slope 2.0(m/s²)/N. Find mass. Predict a at 1.5 N. Explain why plotting driving force instead of net force can produce an offset.

Marzano check - circle: ① ② ③

08 STRETCH: test the model

A 2 kg cart is pulled with 3, 5 and 7 N. Its accelerations are 1, 2 and 3 m/s². Plot pulling force versus acceleration. Use the slope and intercept to infer a constant opposing force.



Marzano check - circle: ① ② ③